



Stony Brook University

DATA ANALYSIS

AMS 572

Inference on one population mean μ

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One population setting

Cross-sectional study: Collect data to test a hypothesis about the mean or median of X

- ▶ Take a random sample from the population.
- ▶ There are different types of sampling schemes. The simplest is the simple random sampling in which every subject in the population has equal chance to be selected.

Statistical inference on one population mean

Suppose we have a random sample of size n : $X_1, X_2, \dots, X_n$ and we wish to draw inference about the population mean μ .

1. Point estimator

- ▶ $\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n} = \frac{\sum_{i=1}^n X_i}{n}$

- ▶ Other estimators: median, mode, trimmed mean

2. Confidence Interval (C.I.)

3. Hypothesis Test

- ▶ Example: μ is the height of adult US males

$$H_0 : \mu \leq 5'6'' \text{ vs } H_1 : \mu > 5'6''$$

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Recap: Normal Distribution

- Probability Density Function (p.d.f.)

X follows normal distribution of mean μ and variance σ^2

$$X \sim N(\mu, \sigma^2)$$

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, -\infty < x < \infty, x \in R$$

$P(a \leq X \leq b) = \int_a^b f(x)dx$ = area under the pdf curve bounded by a and b

- Cumulative Density Function (c.d.f.)

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t)dt$$

$$f(x) = [F(x)]' = \frac{d}{dx}F(x)$$

Recap: Normal Distribution

► Standard Normal Distribution

$$Z \sim N(0, 1)$$

$$X \sim N(\mu, \sigma^2)$$

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

$$\begin{aligned} P(a \leq X \leq b) &= P\left(\frac{a - \mu}{\sigma} \leq \frac{X - \mu}{\sigma} \leq \frac{b - \mu}{\sigma}\right) \\ &= P\left(\frac{a - \mu}{\sigma} \leq Z \leq \frac{b - \mu}{\sigma}\right) \end{aligned}$$

Also known as **Z-score**

Theorem: If two random variables have the same MGF, they have the same distribution.

Normal Distributions: $X \sim N(\mu, \sigma^2)$

$$M_X(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$$

Example: If the mgf of W is $e^{-7t+10t^2}$, what is the distribution of W ?

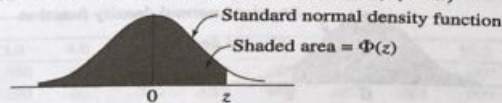


Linear transformation of a normal random variable

► If $X \sim N(\mu, \sigma^2)$ and $Y = aX + b$, a, b are constants, then

$$Y \sim N(a\mu + b, a^2 \sigma^2)$$

Example: Suppose the height of students of AMS 572 is normally distributed. Ten percent of the students are over 6.5 feet tall, while the variance is 0.390625 (or 0.625^2). What is the probability that the height of a student is in between 6 and 7 feet?

Table A.3 Standard Normal Curve Areas $\Phi(z) = P(Z \leq z)$ (cont.)

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9278	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

Distribution of sample mean $\bar{X}$

Theorem: Let $X_1, X_2, \dots, X_n$ be a random sample from a normal population with mean μ , variance σ^2 . Then, the distribution of $\bar{X}$ is

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$



Inference for μ when σ^2 is known under normal distribution



Inference for μ when σ^2 is known under normal distribution

Setup:

- ▶ Assume that the distribution is normal
- ▶ Let $X_1, X_2, \dots, X_n$ be a random sample for a normal distribution with mean μ and variance σ^2 . That is,
 $X \stackrel{iid.}{\sim} N(\mu, \sigma^2), i = 1, \dots, n.$
- ▶ Assume that σ^2 is known.

Inference for μ when σ^2 is known under normal distribution

1. Point estimator for μ

$$\hat{\mu} = \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Note that $E(\hat{\mu}) = E(\bar{X}) = \mu \Rightarrow \hat{\mu} = \bar{X}$ is an unbiased estimator of μ

- $\bar{X}$ is also a maximum likelihood estimator (MLE) and method of moment estimator (MME) of μ .



Inference for μ when σ^2 is known under normal distribution

2 Confidence Interval for μ :

Intuitive approach : $P(c_1 \leq \mu \leq c_2) = 0.95$

$$P(\bar{X} - c_1 \geq \bar{X} - \mu \geq \bar{X} - c_2) = 0.95$$

$$\bar{X} - \mu \sim N(0, \frac{\sigma^2}{n})$$

$$P(\frac{\bar{X} - c_1}{\sigma/\sqrt{n}} \geq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \geq \frac{\bar{X} - c_2}{\sigma/\sqrt{n}}) = 0.95$$

There ARE many ways to choose the c 's.

Pivotal quantity with symmetric pdf, the symmetric CIs are optimal, i.e., they have the shortest lengths for a given confidence level.



Pivotal Quantity (P.Q.) approach for deriving confidence interval

Definition: A pivotal quantity is a function of the sample and parameter of interest whose probability distribution does not depend on the unknown parameters.

Pivotal Quantity (P.Q.) approach for deriving confidence interval

Consider $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$, where σ^2 is known

- ▶ Is $\bar{X}$ a pivotal quantity for μ ?
- ▶ Function of $\bar{X}$ and μ : $\bar{X} - \mu \sim N(0, \frac{\sigma^2}{n})$
- ▶ Another function of $\bar{X}$ and μ : $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$

Derivation for the symmetrical CI's for μ based on pivotal quantity Z

CI for μ , $0 < \alpha < 1$ (e.g. $\alpha = 0.05 \Rightarrow 95\%$ C.I.)

$$P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1 - \alpha$$

$$P(-z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}) = 1 - \alpha$$

$$P(-z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \leq \bar{X} - \mu \leq z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}) = 1 - \alpha$$

$$P(\bar{X} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}) = 1 - \alpha$$

Thus the $100(1 - \alpha)\%$ C.I. for μ is $[\bar{X} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}]$



Confidence Interval Interpretation

- ▶ If we draw 100 different random samples, on average $100(1 - \alpha)\%$ of them will contain μ

Sample $n \longleftrightarrow$ the corresponding $100(1-\alpha)\%$ C.I.

Sample 1, $\bar{x}=5'7'' \longleftrightarrow [5'5'', 5'9'']$

Sample 2, $\bar{x}=5'5'' \longleftrightarrow [5'3'', 5'7'']$

Sample 3, $\bar{x}=5'8'' \longleftrightarrow [5'6'', 5'10'']$

... e.g.) 95% C.I., $\mu=5'7''$, 95% of all these CI's will cover μ

...

Confidence Interval Interpretation

- ▶ For the $100(1-\alpha)\%$ C.I. for μ is $[\bar{X} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}]$, the length of this CI is:

$$L_{sy} = 2 \cdot z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

- ▶ To decrease the length of confidence interval:
 - ▶ increase α , i.e., decrease confidence
 - ▶ increase sample size

Non-symmetric confidence interval

- ▶ Note that $P(-z_{\alpha/3} \leq Z \leq z_{2\alpha/3}) = 1 - \alpha$
- ▶ 100(1- α)% C.I. for μ

$$[\bar{X} - z_{2\alpha/3} \cdot \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/3} \cdot \frac{\sigma}{\sqrt{n}}]$$

The length of this CI is:

$$L_{nsy} = (z_{\alpha/3} + z_{2\alpha/3}) \cdot \frac{\sigma}{\sqrt{n}}$$

- ▶ It can be shown that: $L_{sy} \leq L_{nsy}$
- ▶ Try a few numerical values for α , and see for yourself.

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Sample Size Determination for a z-interval:

On the verbal section of the SAT, the standard deviation is known to be 100. What size sample would we need to construct a 95% confidence interval with a margin of error of 20?

$$n = \left(\frac{(z_{\alpha/2})(\sigma)}{E} \right)^2$$

$$n = \left(\frac{(1.96)(100)}{20} \right)^2 = 96.04$$



Hypothesis testing

- ▶ Null hypothesis H_0
- ▶ Example of null hypothesis for folic acid study:
The incidence of stroke in the current study cohort will be the same as the reported population incidence last year.

Null and Alternative

- ▶ In a test of a hypothesis, we are testing whether some population parameter has a particular value or range
- ▶ For example,

$$H_0 : \mu \leq \mu_0$$

where μ_0 is a known constant

- ▶ The **alternative hypothesis** is complement of null hypothesis

$$H_a : \mu > \mu_0$$

Null and Alternative

- ▶ A law suit analogy:
Example: The OJ Simpson trial
 H_0 : OJ is innocent
 H_a : OJ is guilty

		The truth	
		H_0 : OJ innocent	H_a : OJ guilty
Jury's Decision	H_0	Right decision	Type II error
	H_a	Type I error	Right decision (Power)

Tests of Hypotheses: Seven Steps

1. Design study
2. Establish null hypothesis
3. Determine test statistic to be employed
4. Choose significance level α and establish critical region C_α .
 α is also $P(\text{Type I error})$.
5. Carry out study and collect data
6. Compute statistic from data
7. If statistic is in C_α , reject H_0

.....

Types of hypothesis

One-sided (or one-tailed) test:

- ▶ $H_0 : \mu = \mu_0$ vs $H_a : \mu > \mu_0$
 $H_0 : \mu \leq \mu_0$ vs $H_a : \mu > \mu_0$
- ▶ $H_0 : \mu = \mu_0$ vs $H_a : \mu < \mu_0$
 $H_0 : \mu \geq \mu_0$ vs $H_a : \mu < \mu_0$

Two-sided (or two-tailed) test:

- ▶ $H_0 : \mu = \mu_0$ vs $H_a : \mu \neq \mu_0$



Hypothesis test for $H_0 : \mu = \mu_0$ vs $H_a : \mu > \mu_0$

Setting

- ▶ Data : $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$, σ^2 is known
- ▶ The significance level α (e.g., $\alpha = 0.05$) is given.



Hypothesis test for $H_0 : \mu = \mu_0$ vs $H_a : \mu > \mu_0$

Pivotal Quantity Method

1. Identify a pivotal quantity: Recall that

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \sim N(0, 1) \text{ is a pivotal quantity.}$$

2. The **test statistic** is the pivotal quantity with the value of the parameter of interest under the null hypothesis

Hypothesis test for $H_0 : \mu = \mu_0$ vs $H_a : \mu > \mu_0$

- 3 Derive the decision threshold for your test based on the Type I error rate, that is, the significance level α

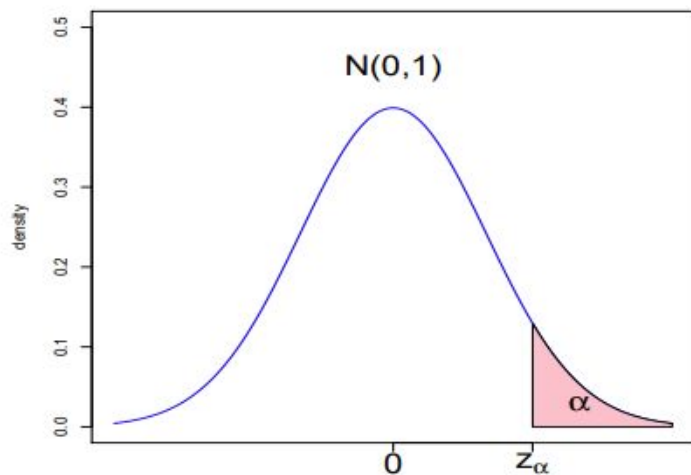
For the pair of hypotheses: $H_0 : \mu = \mu_0$ versus $H_a : \mu > \mu_0$

It is intuitive that one should reject H_0 , in support of the H_a , when the $\bar{X} > \mu_0$. Equivalently, this means to reject H_0 when the test statistic Z_0 is larger than certain positive value c ($Z_0 > c$).

The question is what is the exact value of c , which can be determined based on the significance level α , i.e., how much Type I error we would allow ourselves to commit.

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Hypothesis test for $H_0 : \mu = \mu_0$ vs $H_a : \mu > \mu_0$

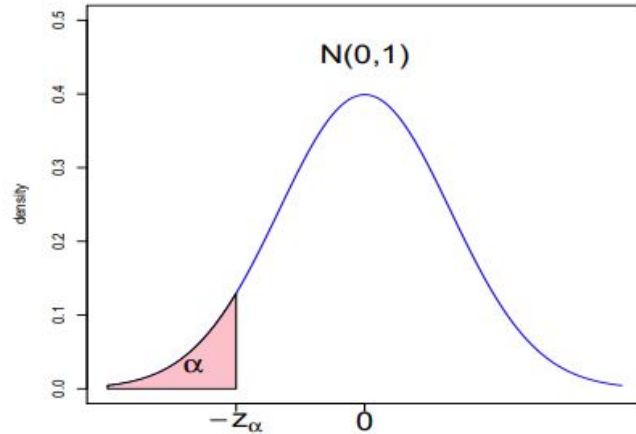


$\therefore$ At the significance level α , we will reject H_0 in favor of H_a if

$$Z_0 \geq z_\alpha$$

Hypothesis test for $H_0 : \mu = \mu_0$ versus $H_a : \mu < \mu_0$

Test statistic: $Z_0 = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1)$



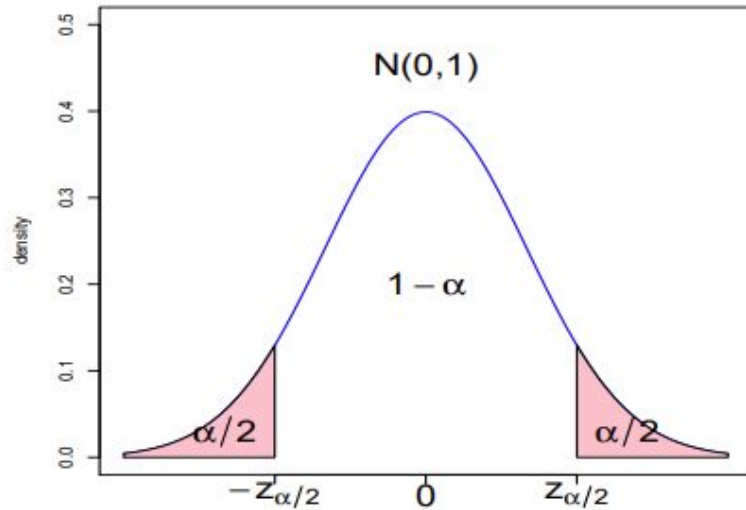
$\therefore$ At the significance level α , we will reject H_0 in favor of H_a if
 $Z_0 \leq -z_\alpha$



Hypothesis test for $H_0 : \mu = \mu_0$ versus $H_a : \mu \neq \mu_0$

Test statistic : $Z_0 = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1)$

Hypothesis test for $H_0 : \mu = \mu_0$ versus $H_a : \mu \neq \mu_0$



P-value

- ▶ We have just discussed the “**rejection/critical region**” approach for decision making.
- ▶ There is another approach for decision making, it is the “**p-value**” approach.

Definition: p-value is the probability that of observing a test statistic value that is as extreme, or more extreme, than the one we observed.

- ▶ Reject H_0 in favor of H_a if $\text{p-value} \leq \alpha$.
For example, for $H_0 : \mu = \mu_0$ versus $H_a : \mu > \mu_0$, the p-value is $P(Z_0 \geq z_0 | H_0)$.

Summary

$H_0 : \mu = \mu_0$ $H_a : \mu > \mu_0$	$H_0 : \mu = \mu_0$ $H_a : \mu < \mu_0$	$H_0 : \mu = \mu_0$ $H_a : \mu \neq \mu_0$
Observed value of test statistic $Z_0 = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \stackrel{H_0}{\sim} N(0, 1)$		
Rejection region : we reject H_0 in favor of H_a at the significance level α if		
$Z_0 \geq z_\alpha$	$Z_0 \leq -z_\alpha$	$ Z_0 \geq z_{\alpha/2}$
p-value = $P(Z_0 \geq z_0 H_0)$	p-value = $P(Z_0 \leq z_0 H_0)$	p-value $= P(Z_0 \geq z_0 H_0)$ $= 2 * P(Z_0 \geq z_0 H_0)$
the area under $N(0,1)$ pdf to the right of z_0	the area under $N(0,1)$ pdf to the left of z_0	twice the area to the right of $ z_0 $



Inference for μ when σ^2 is unknown, large
sample for any distribution

Inference for μ when σ^2 is unknown, large sample for any distribution

Setup:

- ▶ Let $X_1, X_2, \dots, X_n$ be a random sample for a distribution (need not be normal) with mean μ and variance σ^2 .
- ▶ We assume that σ^2 is unknown.
- ▶ Sample size n is large enough ($n \geq 30$)

Theorem: Central Limit Theorem (CLT)

Let $X_1, X_2, \dots, X_n$ be independently and identically distributed (i.i.d.) random variables with common mean μ and variance σ^2 . Then the random variable

$$Y = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

where $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ has a limiting distribution that is normal with mean 0 and variance 1. That is,

$$Y \xrightarrow{d} Z \sim N(0, 1)$$

as $n \longrightarrow \infty$

Slutsky's Theorem

- ▶ If X_n is a sequence of r.v. that converges in distribution to X , and
- ▶ Y_n is a sequence of r.v. that converges in probability to a constant c ,
- ▶ then $W_n = X_n Y_n$ converges in distribution to cX
- ▶ i.e.

$$\lim_{n \rightarrow \infty} \Pr[W_n \leq w] = \Pr[cX \leq w]$$

Theorem: Central Limit Theorem (CLT)

To use the CLT when σ^2 unknown requires *Slutsky's Theorem*.
By both CLT and Slutsky's Theorem,

$$Z = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

is approximately $N(0, 1)$.

Note: X_i 's do not need to be normally distributed

Theorem: Central Limit Theorem (CLT)

To use the CLT when σ^2 unknown requires *Slutsky's Theorem*.
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Note: X_i 's do not need to be normally distributed

CI for μ when σ^2 unknown, large sample for any distribution

- ▶ Since

$$\frac{\bar{X} - \mu}{s/\sqrt{n}} \sim N(0, 1)$$

we have

$$P\left(-z_{\alpha/2} \leq \frac{\bar{X} - \mu}{s/\sqrt{n}} \leq z_{\alpha/2}\right) = 1 - \alpha$$

- ▶ Thus $100(1 - \alpha)\%$ CI for μ is given by

$$\left(\bar{X} - z_{\alpha/2} \frac{s}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{s}{\sqrt{n}}\right)$$

.....

Inference for μ when σ^2 is unknown, large sample for any distribution

The derivations of the hypothesis tests (rejection region and the p-value) are almost the same as the derivation of the exact Z-test in previous set of slides.

Summary

$H_0 : \mu = \mu_0$ $H_a : \mu > \mu_0$	$H_0 : \mu = \mu_0$ $H_a : \mu < \mu_0$	$H_0 : \mu = \mu_0$ $H_a : \mu \neq \mu_0$
Observed value of test statistic $Z_0 = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \stackrel{H_0}{\sim} N(0, 1)$		
Rejection region : we reject H_0 in favor of H_a at the significance level α if		
$Z_0 \geq z_\alpha$	$Z_0 \leq -z_\alpha$	$ Z_0 \geq z_{\alpha/2}$
p-value = $P(Z_0 \geq z_0 H_0)$	p-value = $P(Z_0 \leq z_0 H_0)$	p-value = $P(Z_0 \geq z_0 H_0)$ = $2 * P(Z_0 \geq z_0 H_0)$
the area under $N(0, 1)$ pdf to the right of z_0	the area under $N(0, 1)$ pdf to the left of z_0	twice the area to the right of $ z_0 $



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 $X \stackrel{iid.}{\sim} N(\mu, \sigma^2), i = 1, \dots, n.$
- ▶ Assume that σ^2 is unknown.
- ▶ Assume sample size n is small.

Inference for μ when σ^2 is unknown, small sample for normal distribution

Under this scenario, the distribution of

$$\frac{\bar{X} - \mu}{s/\sqrt{n}}$$

is not normal.



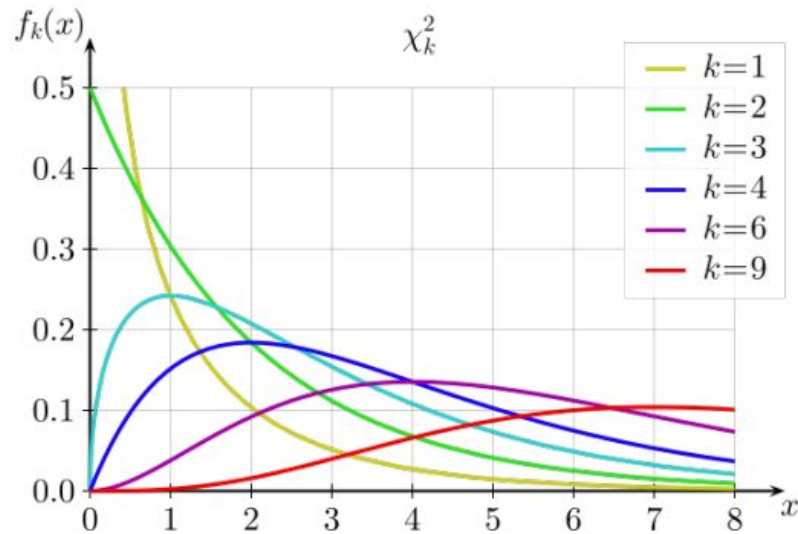
- ▶ If a random variable X is normally distributed with mean μ and variance σ^2 , then for a random sample of size n , the quantity

$$\frac{(n-1)s^2}{\sigma^2}$$

has a chi-square distribution with $n-1$ degrees of freedom, which we denote by χ_{n-1}^2

- ▶ Let $Z_1, Z_2, \dots, Z_k \stackrel{i.i.d.}{\sim} N(0, 1)$, then $W = \sum_{i=1}^k Z_i^2 \sim \chi_k^2$
- ▶ chi-square distribution is a special gamma distribution

χ^2 Distribution



https://en.wikipedia.org/wiki/Chi-squared_distribution

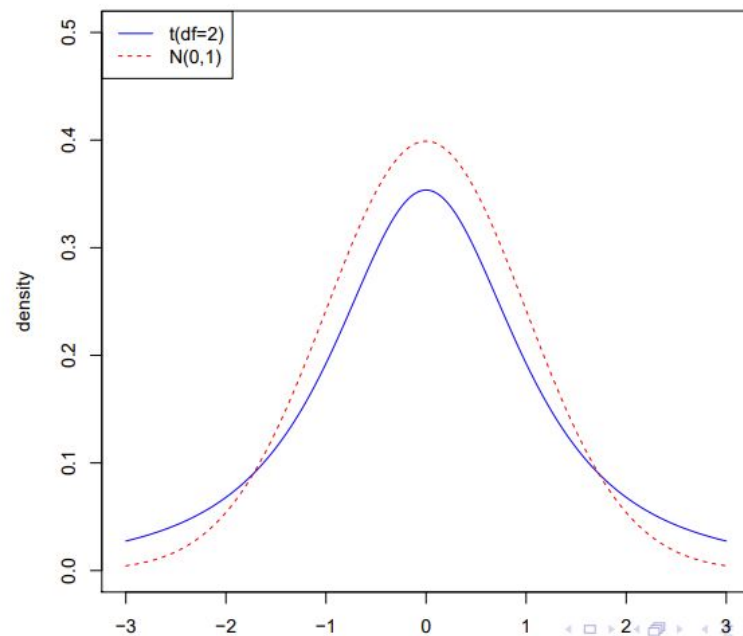
- ▶ Let $Z \sim N(0, 1)$ and $W \sim \chi^2_\nu$
- ▶ If Z and W are independent, then

$$T = \frac{Z}{\sqrt{W/\nu}}$$

will follow the t -distribution with ν degrees of freedom.

- ▶ t -distribution has heavier tails than normal distribution

t Distribution



Theorem: Sampling from the normal population

Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma^2)$, then

1. $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$
2. $W = \frac{(n-1)s^2}{\sigma^2} \sim \chi_{n-1}^2$
3. $\bar{X}$ and s^2 (and thus W) are independent.

Thus we have

$$T = \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t_{n-1}$$

CI for μ when σ^2 unknown, small sample for normal distribution

- Since

$$\frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t_{n-1}$$

we have $P\left(-t_{n-1,\alpha/2} \leq \frac{\bar{X} - \mu}{s/\sqrt{n}} \leq t_{n-1,\alpha/2}\right) = 1 - \alpha$

- Thus $100(1 - \alpha)\%$ CI for μ is given by

$$\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$$

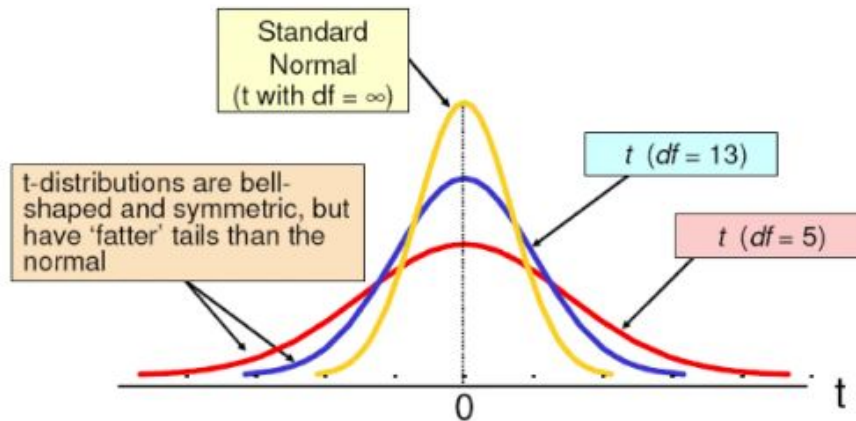
Summary

$H_0 : \mu = \mu_0$ $H_a : \mu > \mu_0$	$H_0 : \mu = \mu_0$ $H_a : \mu < \mu_0$	$H_0 : \mu = \mu_0$ $H_a : \mu \neq \mu_0$
Observed value of test statistic $T_0 = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \stackrel{H_0}{\sim} t_{n-1}$		
Rejection region : we reject H_0 in favor of H_a at the significance level α if		
$T_0 \geq t_{n-1,\alpha}$	$T_0 \leq -t_{n-1,\alpha}$	$ T_0 \geq t_{n-1,\alpha/2}$
p-value = $P(T_0 \geq t_0 H_0)$	p-value = $P(T_0 \leq t_0 H_0)$	p-value = $P(T_0 \geq t_0 H_0) = 2 \cdot P(T_0 \geq t_0 H_0)$
the area under t_{n-1} pdf to the right of t_0	the area under t_{n-1} pdf to the left of t_0	twice the area under t_{n-1} to the right of $ t_0 $

.....



Student's t Distribution...



Note: t dist'n approaches z dist'n as n increases

Assessing Normality

- ▶ How do we assess whether the normal distribution model is a reasonable fit for a particular set of data?
- ▶ One graphical approach: quantile-quantile (QQ) plot
- ▶ Plot quantiles of the observed data distribution versus the quantiles of the normal distribution, i.e we plot the pairs

$$\left(\Phi^{-1} \left(\frac{i - 0.5}{n} \right), x_{(i)} \right), i = 1, \dots, n$$

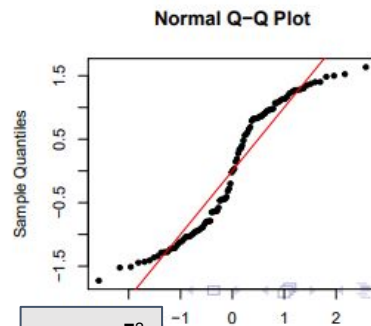
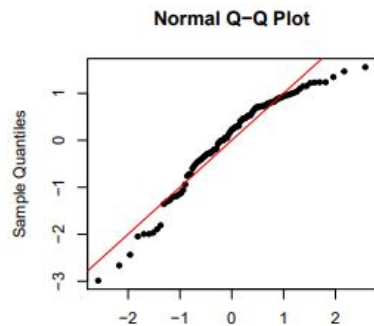
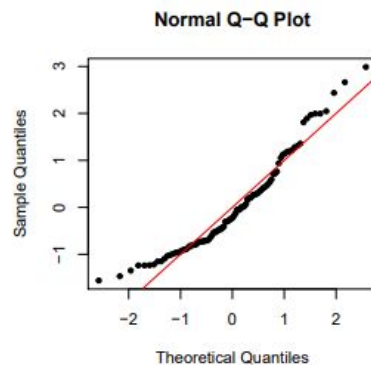
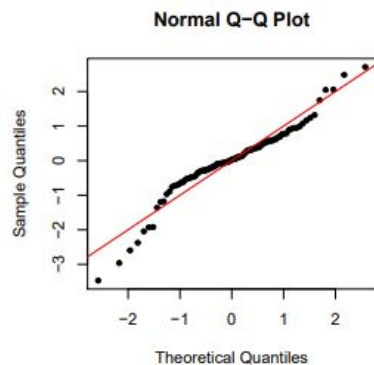
where $x_{(i)}$'s are the order statistics

- ▶ Straight line indicates normality assumption reasonable

QQ plot in R

```
> x <- rnorm(100)
> qqnorm(x)
> qqline(x)
```

QQ plot: Examples where normal distribution assumption is violated





Assessing Normality

- ▶ Alternatively, statistical tests for univariate normality include Shapiro-Wilk, Kolmogorov-Smirnov, Lilliefors, and Anderson-Darling tests.
- ▶ Razali et al. (2011) (Journal of Statistical Modeling and Analytics) concluded that Shapiro-Wilk test has the best power for a given significance among these tests.

Shapiro Wilk test

- ▶ Test statistic

$$W = \frac{(\sum_{i=1}^n a_i x_{(i)})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

where $x_{(i)}$'s are the order statistics,

$$(a_1, \dots, a_n) = \mathbf{m}^T V^{-1} / (\mathbf{m}^T V^{-1} V^{-1} \mathbf{m}^T)^{1/2}$$

$\mathbf{m} = (m_1, \dots, m_n)^T$, $m_1, \dots, m_n$ and V are the expected values and covariance matrix of the order statistics from i.i.d $N(0, 1)$, respectively.

- ▶ The empirical distribution of W is obtained via Monte Carlo simulations.

Shapiro Wilk test in R

The underlying null hypothesis of Shapiro Wilk test is that data came from a Normal Distribution.

```
> x <- rnorm(100)  Generating data from Normal Distribution  
> shapiro.test(x)
```

Shapiro-Wilk normality test

```
data: x  
W = 0.99273, p-value = 0.8713
```

```
> x <- rexp(100)  Generating data from Exponential Distribution  
> shapiro.test(x)
```

Shapiro-Wilk normality test

```
data: x  
W = 0.82652, p-value = 1.763e-09
```

A higher W value indicates that your data closely follows a normal distribution. It ranges from 0 to 1, with values nearer to 1 suggesting normality.



A higher W value indicates that your data closely follows a normal distribution. It ranges from 0 to 1, with values nearer to 1 suggesting normality.

The p-value tells us whether to reject the null hypothesis (that the data is normally distributed). A common threshold is 0.05:

- If the p-value is greater than 0.05, we fail to reject the null hypothesis, suggesting the data is normally distributed.
- If the p-value is less than or equal to 0.05, we reject the null hypothesis, indicating the data does not follow a normal distribution.

Example 1: Jerry is planning to purchase a sports good store. He calculated that in order to cover basic expenses, the average daily sales must be greater than \$525.

Scenario A. He checked the daily sales of 36 randomly selected business days, and found the average daily sales to be \$565 with a standard deviation of \$150.

Scenario B. Now suppose he is only allowed to sample 9 days. And the 9 days sales are \$510, 537, 548, 592, 503, 490, 601, 499, 640.

For A and B, determine if Jerry can conclude the daily sales to be greater than \$525 at the significance level of $\alpha = 0.05$. What is the p-value for each scenario?

Example 2 (Recap): Let $X_1, X_2, \dots, X_n$ be i.i.d random sample of size n ($n \geq 30$) from a population with unknown and non-normal distribution. Derive the $100(1 - \alpha)\%$ CI for μ .

Ans: According to the CLT and Slutsky's Theorem,

$$Z = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} \sim N(0, 1),$$

$$\text{where } s = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}}.$$

$$\text{Since } P\left(-z_{\frac{\alpha}{2}} \leq \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} \leq z_{\frac{\alpha}{2}}\right) = 1 - \alpha,$$

then the $100(1 - \alpha)\%$ CI for μ is $(\bar{X} - z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}}, \bar{X} + z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}})$.

*If the population variance σ^2 is known, then the CI will be

$$(\bar{X} - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}).$$

Example 3 (Recap): Let $X_1, X_2, \dots, X_n$ be i.i.d random sample of size n from $N(\mu, \sigma^2)$, where σ^2 is unknown. Derive the $100(1 - \alpha)\%$ CI for μ .

Ans: Since $\frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} \sim t_{n-1}$, then

$$P\left(-t_{n-1, \frac{\alpha}{2}} \leq \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} \leq t_{n-1, \frac{\alpha}{2}}\right) = 1 - \alpha,$$

so the $100(1 - \alpha)\%$ CI for μ is

$$\left(\bar{X} - t_{n-1, \frac{\alpha}{2}} \frac{s}{\sqrt{n}}, \bar{X} + t_{n-1, \frac{\alpha}{2}} \frac{s}{\sqrt{n}}\right).$$

...